

pfSense Frontend Practice – Hands-On Assignment & Assessment Document

LAB 1 – Networking & Interface Configuration:

```
options=0
maxupd: 128 defer: off version: 1400
syncok: 1
groups: pfsync
[2.8.1-RELEASE][root@pfSense.home.arpal/root: exit
exit
VirtualBox Virtual Machine - Netgate Device ID: baf42031116a00edf81d

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WANUPLINK (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
                               v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe9a:c39b/64
LANCORE (lan)   -> em1 -> v4: 192.168.10.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █

Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20251126-2112
Bootup complete

FreeBSD/amd64 (pfSense.home.arpal) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: baf42031116a00edf81d

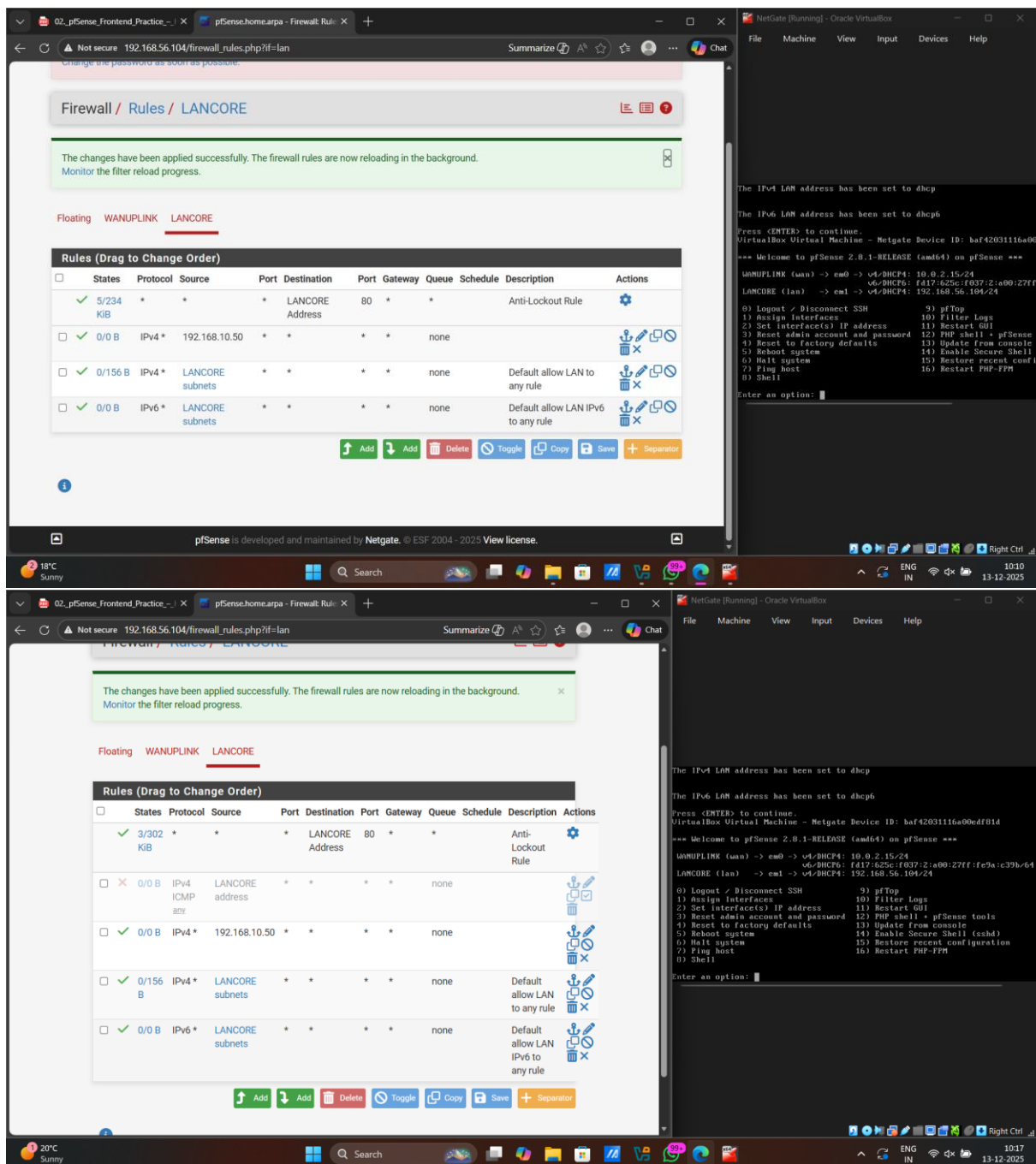
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WANUPLINK (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
                               v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe9a:c39b/64
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6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █
```

LAB 2 – Firewall Rule Logic & Policy Enforcement:



02_pfsense_frontend_practice_... X

pfsense.home.arpa - Firewall Sch: X

+

192.168.56.104/firewall_schedule.php

Summarize

Chat

WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Firewall / Schedules

Schedules

Name	Range: Date / Times / Name	Description	Actions
WorkingHours	Mon - Fri / 9:00-18:00 /	timeframe for students	

Indicates that the schedule is currently active.

+ Add

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NetGate [Running] - Oracle VirtualBox

File Machine View Input Devices Help

7) Ping host
8) Shell

Enter an option: 8

(Z.0.1-RELEASE)[root@pfsense.home.arpa:/root]: ping 0.0.0.0
PING 0.0.0.0 (0.0.0.0): 56 data bytes
64 bytes from 0.0.0.0: icmp_seq=0 ttl=255 time=51.541 ms
64 bytes from 0.0.0.0: icmp_seq=1 ttl=255 time=44.504 ms
64 bytes from 0.0.0.0: icmp_seq=2 ttl=255 time=40.263 ms
64 bytes from 0.0.0.0: icmp_seq=3 ttl=255 time=30.247 ms
--- 0.0.0.0 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 30.247/43.734/51.541/5.046 ms
(Z.0.1-RELEASE)[root@pfsense.home.arpa:/root]: ping 0.0.0.0
PING 0.0.0.0 (0.0.0.0): 56 data bytes
64 bytes from 0.0.0.0: icmp_seq=0 ttl=255 time=63.870 ms
64 bytes from 0.0.0.0: icmp_seq=1 ttl=255 time=51.643 ms
64 bytes from 0.0.0.0: icmp_seq=2 ttl=255 time=26.179 ms
--- 0.0.0.0 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 26.179/47.233/63.870/15.703 ms
(Z.0.1-RELEASE)[root@pfsense.home.arpa:/root]:

02_pfsense_frontend_practice_... X

pfsense.home.arpa - Status: Sys: X

+

192.168.56.104/status_logs_filter.php

Summarize

Chat

Action	Time	Interface	Rule	Source	Destination	Protocol
X	Dec 9 15:26:28	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:28	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:28	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:30	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:30	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:30	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:30	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:30	WANUPLINK	(12013)	172.16.30.190:54672	172.16.31.255:51007	UDP
X	Dec 9 15:26:30	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:31	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:33	WANUPLINK	Default deny rule IPv6 (1000000105)	[fe80::a00:27ff:fe4f:e16]	[ff02::1:861:20ce]	ICMPv6
X	Dec 9 15:26:33	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:34	WANUPLINK	(12013)	172.16.30.190:54672	172.16.31.255:51007	UDP
X	Dec 9 15:26:34	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:34	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:34	WANUPLINK	(12013)	172.16.1.41:2190	172.16.255.255:2190	UDP
X	Dec 9 15:26:35	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP
X	Dec 9 15:26:36	WANUPLINK	Default deny rule IPv6 (1000000105)	[fe80::a00:27ff:fe4f:e16]	[ff02::2:ff18:6120]	ICMPv6
X	Dec 9 15:26:37	WANUPLINK	(11005)	0.0.0.0:68	255.255.255.255:67	UDP

NetGate [Running] - Oracle VirtualBox

File Machine View Input Devices Help

--- 0.0.0.0 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 26.179/47.233/63.870/15.703 ms
(Z.0.1-RELEASE)[root@pfsense.home.arpa:/root]: exit
VirtualBox Virtual Machine - Netgate Device ID: haf42631116a09cdf81d
*** Welcome to pfsense 2.0.1-RELEASE (amd64) on pfsense ***
WANUPLINK (wan) -> em0 -> v4/BHCP4: 10.0.2.15/24
v6/BHCP6: fd17:625c:f037:2:a00:27ff:fe9a:c3
LANCORE (lan) -> em1 -> v4/BHCP4: 192.168.56.104/24
0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pTop
10) Filter Logs
11) Restart GUI
12) PHP shell - pfsense tools
13) Update from console
14) Enable Secure Shell (ssh)
15) Restore recent configuration
16) Restart PHP-FPM
Enter an option:

LAB 3 – NAT & Port Forwarding:

02_pfsense_frontend_practice_... x

pfsense.home.arpa - Firewall NAT

+

192.168.56.104/firewall_nat_out.php

Summarize

Not secure

Outbound NAT Mode

Mode

☐

Automatic outbound NAT rule generation. (IPsec passthrough included)

☒Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)

☐Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)

☐Disable Outbound NAT rule generation. (No Outbound NAT rules)

Save

Mappings

Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description	Actions
☐	WANUPLINK	192.168.10.0/24	*	*	*	WANUPLINK address	*		outbound Nat hybrid

AddAddDeleteToggleSave

Automatic Rules

Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description
☒	WANUPLINK	127.0.0.0/8 ::1/128	*	*	500	WANUPLINK address	*	Auto created rule for ISAKMP
☒	WANUPLINK	127.0.0.0/8 ::1/128	*	*	*	WANUPLINK address	*	Auto created rule

i

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1041 13-12-2025

NetGate (Running) - Oracle VirtualBox

File Machine View Input Devices Help

VirtualBox Virtual Machine - Netgate Device ID: baf420

8.0.8.0 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet round-trip min/avg/max/stddev = 26.179/47.233/63.878/12.8.1-RELEASE[[root@pfsense.home.arpa]/root: exit
exit]
Welcome to pfsense 2.0.1-RELEASE (amd64) on pfSense
WANUPLINK (wan) -> em0 -> v4/BHCP4: 10.0.2.15/24
LANCORE (lan) -> em1 -> v4/BHCP4: 192.168.56.104/24
0) Logout / Disconnect SSH 9) pfTop
1) Assign Interfaces 10) Filter Logs
2) Set interface(s) IP address 11) Restart GUI
3) Reset admin account and password 12) PHP shell +
4) Reset to factory defaults 13) Update from
5) Reboot system 14) Enable Secur
6) Halt system 15) Restore race
7) Ping host 16) Restart PHP
8) Shell
Enter an option:

02_pfsense_frontend_practice_... x

pfsense.home.arpa - Firewall NAT

+

192.168.56.104/firewall_nat_out.php

Summarize

Not secure

The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Firewall / NAT / Outbound

Port Forward 1:1 Outbound NPT

Outbound NAT Mode

Mode

☐

Automatic outbound NAT rule generation. (IPsec passthrough included)

☒Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)

☐Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)

☐Disable Outbound NAT rule generation. (No Outbound NAT rules)

Save

Mappings

Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description	Actions
☐	WANUPLINK	192.168.10.0/24	*	*	*	WANUPLINK address	*		outbound Nat hybrid

AddAddDeleteToggleSave

Automatic Rules

Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description
☒	WANUPLINK	127.0.0.0/8 ::1/128	*	*	500	WANUPLINK address	*	Auto created rule for ISAKMP
☒	WANUPLINK	127.0.0.0/8 ::1/128	*	*	*	WANUPLINK	*	Auto created rule

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Search

1037 13-12-2025

NetGate (Running) - Oracle VirtualBox

File Machine View Input Devices Help

VirtualBox Virtual Machine - Netgate Device ID: baf420

8.0.8.0 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet round-trip min/avg/max/stddev = 26.179/47.233/63.878/12.8.1-RELEASE[[root@pfsense.home.arpa]/root: exit
exit]
Welcome to pfsense 2.0.1-RELEASE (amd64) on pfSense
WANUPLINK (wan) -> em0 -> v4/BHCP4: 10.0.2.15/24
LANCORE (lan) -> em1 -> v4/BHCP4: 192.168.56.104/24
0) Logout / Disconnect SSH 9) pfTop
1) Assign Interfaces 10) Filter Logs
2) Set interface(s) IP address 11) Restart GUI
3) Reset admin account and password 12) PHP shell +
4) Reset to factory defaults 13) Update from
5) Reboot system 14) Enable Secur
6) Halt system 15) Restore race
7) Ping host 16) Restart PHP
8) Shell
Enter an option: